```
!pip install gensim
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_2
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-pac
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 64.8 MB/s eta 0:0
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from gensim.models import KeyedVectors
from gensim.scripts.glove2word2vec import glove2word2vec
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.decomposition import PCA

# Define file paths
raw_glove_path = 'glove.6B.100d.txt'
w2v_glove_path = 'glove.6B.100d.w2v.txt'

# Download and extract if unavailable (Standard Colab Procedure)
if not os.path.exists(w2v_glove_path):
    print("Fetching GloVe embeddings...")
    os.system('wget http://nlp.stanford.edu/data/glove.6B.zip')
    os.system('unzip glove.6B.zip')

# Transform format
print("Formatting embeddings for Gensim...")
glove2word2vec(raw_glove_path, w2v_glove_path)

# Initialize the embedding model
print("Loading model into memory...")
embedding_model = KeyedVectors.load_word2vec_format(w2v_glove_path, binary=
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/tmp/ipykernel_4282/3356179966.py:23: DeprecationWarning: Call to deprecat
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